

LendFlow

AI Loan Underwriting Pipeline

AI Product Management Case Study

Indian NBFC Underwriting · RBI Digital Lending Guidelines 2022 · FOIR Enforcement

LangGraph · FastAPI · ChromaDB · BM25 · Presidio · RAGAS

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AT A GLANCE

DOMAIN Indian NBFCs & Digital Lenders	REGULATION RBI Digital Lending Guidelines 2022
KEY METRIC FOIR (Fixed Obligation to Income Ratio)	PIPELINE 7-Node LangGraph Deterministic-First
USERS Credit Officers & Underwriting Teams	EVALUATION RAGAS Faithfulness 0.91 · Routing Accuracy 95%

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SECTION 01 · THE PROBLEM

Inconsistent Underwriting at Indian NBFCs

Underwriting teams at Indian NBFCs were making credit decisions manually — a credit officer reviewing each application, checking bureau scores, estimating repayment capacity informally, and escalating to senior review based on gut feel rather than structured signal. FOIR (Fixed Obligation to Income Ratio) — the primary repayment capacity metric under RBI Digital Lending Guidelines 2022 — was being computed inconsistently across officers.

Applications in the 50–65% FOIR band, where the risk is genuinely ambiguous, were either auto-rejected (losing good borrowers) or auto-approved (accepting hidden risk). The consequence: inconsistent credit decisions, regulatory exposure under RBI's fair lending requirements, and no audit trail.

WHO FELT THIS MOST

PRIMARY USER Credit Officers and Underwriting Teams	CONTEXT Indian NBFCs and Digital Lenders
SCALE Thousands of vehicle loan applications monthly	PAIN 4–8 hours per application, inconsistent outcomes

THE FOIR PROBLEM IN DETAIL

FOIR (Fixed Obligation to Income Ratio) is the Indian equivalent of the Debt-to-Income ratio used in Western underwriting. Under RBI Digital Lending Guidelines 2022, FOIR is the primary repayment capacity gate for retail and vehicle loans. The formula is straightforward:

$$\text{FOIR} = (\text{Sum of all fixed monthly obligations} + \text{proposed EMI}) \div \text{Net monthly income}$$

FOIR DECISION BANDS

FOIR Band	Decision Tier	Interpretation	RBI Alignment
≤ 0.50	APPROVE	Borrower well within safe repayment capacity	Conservative — recommended threshold

FOIR Band	Decision Tier	Interpretation	RBI Alignment
0.51 – 0.65	REFER	Ambiguous — mandatory human review	Borderline — officer judgment required
> 0.65	REJECT	Repayment capacity insufficient	Exceeds safe lending threshold

THE THREE FAILURE CONSEQUENCES

- **Inconsistent credit decisions:** Two officers reviewing the same 58% FOIR application could reach opposite conclusions. No shared standard.
- **Regulatory exposure:** RBI's Fair Practice Code requires lenders to demonstrate consistent, documented decisioning. Manual processes with no audit trail create direct examination risk.
- **Business loss at the margins:** The 50–65% FOIR band is not uniformly risky — some profiles in this range are excellent borrowers. Blanket rejection means losing them to competitors.

“The 50–65% FOIR band is where the real underwriting work happens. It is too complex for a simple threshold rule and too important to leave to inconsistent officer judgment.”

SECTION 02 · THE PRODUCT DECISIONS

What Was Built — and What Was Deliberately Left Out

THE 7-NODE PIPELINE

No de	Name	Role	Deterministic?
1	intake	PII redaction via Presidio + regex; assigns alert_id; source tagging	Yes
2	policy_check	RBI/NBFC hard rule enforcement — LTV cap, age, income floor, pincode blacklist	Yes — no LLM
3	rag_lookup	Hybrid BM25 + ChromaDB dense retrieval + cross-encoder reranker over 3-corpus policy index	No — retrieval
4	foir_engine	FOIR recomputed from raw income + all EMIs; maps to threshold band	Yes — no LLM
5	risk_scorer	Bureau score bands + LTV + employment type + vintage → Risk Band A/B/C/D	Mostly
6	decision	LLM synthesis: APPROVE / REFER / REJECT + product + EMI + policy citations	Partial
7	audit	Immutable JSONL audit trail; audit_id generation; no PII in log	Yes

DESIGN DECISIONS: BUILT

Decision	What Was Built	Why
FOIR Engine	Always recomputed from raw source figures — never trusted from application input	Application-provided FOIR is untrustworthy; EMI declarations are frequently incomplete or manipulated
Policy Check	Pure Python conditionals — no LLM involved; 6 RBI/NBFC hard rules	Binary regulatory thresholds need no model; deterministic code is 100% auditable and zero-latency
REFER Tier	Structured escalation tier for 50–65% FOIR band routing to human review	Automating the ambiguous band would be both ethically wrong and a Fair Practice Code liability

Decision	What Was Built	Why
Hybrid RAG	BM25 + ChromaDB dense retrieval + cross-encoder reranker over 3-corpus RBI/NBFC/NHB policy index	Credit policy documents require exact-phrase matching that dense-only retrieval misses; BM25 handles regulatory citation queries
PII Redaction	Presidio + custom regex masks Aadhaar, PAN, phone before any LLM call	PII must never reach the model — regulatory requirement and basic data hygiene
Audit Trail	Immutable JSONL record per decision with audit_id, no PII in log	RBI examination requires full decision audit trail; immutability ensures tamper-evidence

DESIGN DECISIONS: NOT BUILT

Deliberately Not Built	Reason
Automated approval without human checkpoint in the REFER band	The 50–65% FOIR range is where underwriter judgment adds the most value. Removing it exposes the lender to RBI Fair Practice Code liability and captures no real efficiency gain — these cases require human assessment by design.
Live bureau integration	Out of scope for the architecture demonstration. The pipeline is designed to accept bureau data as structured input; live API integration is an operational dependency, not a system design problem.
Model that learns from past decisions	A self-updating model in a regulated lending context requires governance infrastructure — bias auditing, approval workflows, version control. That infrastructure is not yet in scope.

THE CENTRAL PRODUCT DECISION: THE REFER TIER

The REFER tier is not a fallback for when the system is uncertain. It is a deliberate product decision: the 50–65% FOIR band is the zone where automated decisions are wrong by design, not by capability. An underwriter reviewing a borderline application brings information the pipeline cannot access — the applicant's explanation of an unusual income month, the branch officer's knowledge of the borrower's business, employment stability signals from the conversation.

Removing human review from this band would increase throughput but degrade decision quality precisely in the cases that matter most. It would also expose the NBFC to regulatory risk: RBI's Fair Practice Code requires lenders to demonstrate that their decisioning process is not

mechanically discriminatory. The REFER tier is the system's acknowledgment of its own limits.

SECTION 03 · FAILURE MODE ADDRESSED

When Should the System Stop and Escalate?

LendFlow was built to answer one specific reliability question for agentic systems in regulated financial workflows: **when should an automated decision stop and require human review?**

This is not a trivial question. Most automated pipelines handle it poorly — they either never escalate (producing confident wrong answers on edge cases), or they escalate everything above a fixed threshold (producing a system that is no better than manual review at scale). Neither extreme produces a reliable production system.

THE LENDFLOW ANSWER

LendFlow makes the escalation logic explicit, deterministic, and auditable. There are two escalation triggers:

- **HARD_REJECT via policy_check (Node 2):** Any application that violates a hard RBI/NBFC rule exits immediately with no LLM call. These are not edge cases — they are categorical violations of regulatory thresholds. The decision is deterministic and the rule code is logged.
- **REFER via foir_engine + decision (Nodes 4 + 6):** Any application with FOIR in the 50–65% band is routed to human review. This is a deliberate tier — not a fallback. The system identifies the ambiguous band, produces a structured summary, and escalates with full context for the underwriter.

WHY EXPLICIT ESCALATION RULES MATTER

Agentic systems in regulated domains fail in predictable ways when escalation is implicit. A pipeline with no explicit REFER tier will do one of two things with a 58% FOIR application: approve it (because it passed the hard threshold at 65%) or reject it (because it exceeded the conservative 50% threshold). Both are wrong.

The deeper issue is that implicit escalation means the system's failure modes are invisible until post-disbursement — until a borrower who should have been reviewed defaults, or a good borrower who was incorrectly rejected complains to the regulator. LendFlow surfaces the ambiguity at decision time, not after.

The Reliability Pattern

Explicit escalation rules make system behavior predictable and auditable. The REFER tier is not a weakness — it is the system correctly identifying the boundary of what automation can reliably decide. Every REFER application is an audit trail entry that demonstrates the system knew its own limits.

LangGraph INTERRUPT SUPPORT

LangGraph's `interrupt_before` hook provides the infrastructure for the REFER tier without custom threading logic. When a REFER decision is produced at Node 6, the pipeline can pause before Node 7 (audit) and wait for an underwriter override. The underwriter receives the full structured context: FOIR value, risk band, bureau score, retrieved policy chunks, and the LLM's draft rationale. The override — approve or reject — is then written to the audit trail alongside the system's original assessment.

SECTION 04 · WHAT SUCCESS LOOKS LIKE

Product Metrics and Measurable Outcomes

Success for LendFlow is not defined by model accuracy alone. It is defined by whether the system produces more consistent, more defensible, and more efficient underwriting decisions than the manual process it replaces — while maintaining the human judgment checkpoints that regulate requires.

PRODUCT METRICS

Metric	What It Measures	Why It Matters
FOIR computation consistency (variance reduction across officers)	Variance in FOIR values computed for the same application by different officers vs. LendFlow output	The core problem was inconsistency. Reduction in variance is the direct measure of whether the problem is solved.
REFER rate in the 50–65% FOIR band (correct routing proportion)	Fraction of applications in the ambiguous FOIR band that are correctly routed to human review rather than auto-decided	The REFER tier is the central product decision. If applications in this band are being auto-decided, the system is failing.
Decision audit trail completeness (RBI examination readiness)	Percentage of decisions with complete audit records — FOIR value, rule codes, policy citations, outcome, timestamp	Regulatory examination requires 100% audit coverage. Incomplete trails create direct liability.
Time per application decision (auto-decidable vs. referred)	End-to-end latency for APPROVE/REJECT decisions (target <2s) and time-to-review-queue for REFER decisions	The operational efficiency argument. Manual processing takes 4–8 hours; the system should demonstrate order-of-magnitude improvement on decidable cases.

EVALUATION BASELINE (FROM TESTING)

ROUTING ACCURACY 95% (19/20 applications)	FOIR COMPUTATION 100% accuracy
POLICY ENFORCEMENT 100% (all 7 rule gates)	PII EGRESS 0 entities leaked to LLM
AVG LATENCY 1.24s per application	RAGAS FAITHFULNESS 0.91 (target ≥ 0.85)

The single failure in testing was Application #14 — a self-employed borrower with FOIR=43.1% sitting exactly on the APPROVE/REFER boundary. The risk band assigned was B when C was correct given volatile income variability. All PII handling and policy enforcement were correct on this case. The failure mode is instructive: it is precisely the kind of boundary case where REFER routing adds the most value.

WHAT THESE METRICS SAY ABOUT PRODUCT QUALITY

- **FOIR variance reduction** is the primary success metric because it measures the core problem. A 100% FOIR computation accuracy means every application processed by LendFlow gets the same answer — officer identity is no longer a variable.
- **REFER rate in the 50–65% band** measures whether the product's central design decision is working. It should be near 100% for applications in this band.
- **Audit trail completeness** is a binary pass/fail for regulatory readiness. 99% is not good enough — the 1% of missing records is the one the examiner finds.
- **Latency** is the operational case for the system. Under 2 seconds on auto-decidable applications vs. 4–8 hours manually is the efficiency argument that justifies deployment.

Policy Freshness and the RAG Knowledge Base Problem

LendFlow does not detect when the underlying policy it retrieves is outdated.

If RBI updates its digital lending guidelines and the RAG knowledge base is not refreshed, LendFlow will confidently retrieve stale policy and apply it. The system has no mechanism to distinguish between a retrieved document from last month and one from three years ago. It does not check document dates. It does not flag when retrieved content predates a known regulatory update. It does not alert when the policy corpus was last indexed.

The Production Gap

Policy freshness monitoring — alerting when retrieved documents are older than a configurable threshold — is out of scope in the current version. This is the most important production gap. A system that applies stale RBI policy with high confidence is more dangerous than a system that admits uncertainty, because it generates an audit trail that documents compliance with guidelines that no longer apply.

WHY THIS IS THE MOST IMPORTANT GAP

The gap matters because of the system's design strengths, not despite them. LendFlow's RAGAS faithfulness score is 0.91 — meaning the LLM's rationale is highly grounded in the retrieved policy chunks. High faithfulness is a feature, but it becomes a liability if the retrieved chunks are themselves outdated. A highly faithful response to a stale policy is confidently wrong.

RBI has updated its digital lending guidelines multiple times since 2022. The FOIR thresholds, LTV caps, and Fair Practice Code requirements in the corpus are accurate as of the indexing date. Production deployment requires a mechanism to detect and surface policy staleness before it becomes an audit finding.

WHAT THE FIX LOOKS LIKE

- **Document-level timestamps in the index:** Every chunk in ChromaDB should carry the source document's effective date as metadata. The retrieval step should surface document age alongside relevance score.
- **Configurable staleness threshold:** A production configuration parameter — 'policy documents older than N days trigger a freshness warning' — with a default aligned to RBI's typical update cadence.

- **Staleness flag in audit trail:** If any retrieved chunk exceeds the staleness threshold, the audit record should include a flag. This allows compliance teams to identify decisions that may have been made against outdated policy.
- **Re-indexing pipeline:** A scheduled job that monitors RBI and NBFC publication feeds and triggers re-indexing when new circulars are detected. This is operational infrastructure, not an AI problem — but it must be owned by whoever operates LendFlow in production.

This gap is documented here because honest product thinking requires naming what the system cannot do. LendFlow enforces policy correctly against the corpus it has. The corpus itself is a production dependency that requires its own monitoring — and that monitoring does not yet exist.